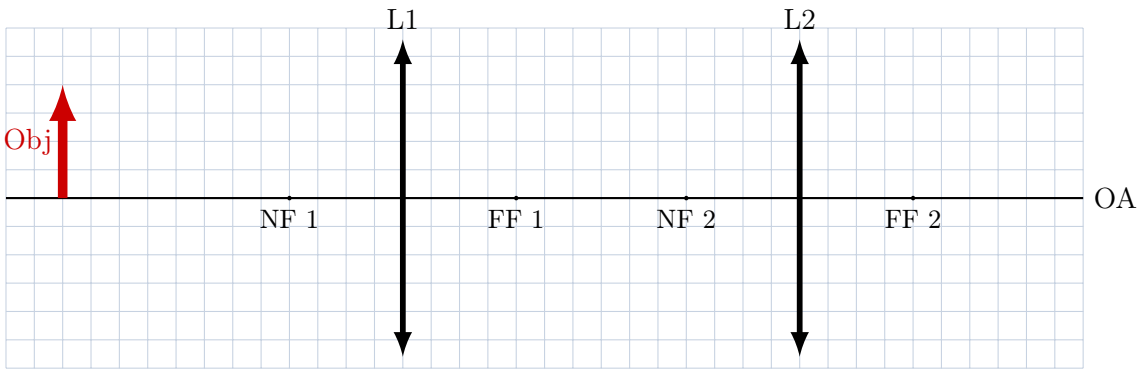
1. Two converging lenses, each with $f = 4$ cm, are 14 cm apart. A 4 cm tall object stands 12 cm in front of the first lens. On the grid, one small square is 1 cm.



A. Use ray tracing to locate the first and second images. Mark and label each one.

B. Use the thin-lens equation to find d_{i1} and d_{i2}.

C. Find M_1, M_2 and the system magnification. How tall is the final image? Is it upright or inverted?

2. A star's rays come in 0.02 rad off the OA. You build a telescope with an objective of $f_o = 30$ cm and an eyepiece of $f_e = 5$ cm.

A. How far apart should you place the two lenses?

B. Sketch the two lenses and trace the rays through the telescope. Mark where the focal points coincide.

C. What is the angular magnification of the telescope?

D. How big is the angle θ_e subtended in the eyepiece?